

SQL LAB 1

Parth Amol Surve, 59, D6

17/08/2026

Aim: To study DDL and DDL commands.

Desc customers;

MySQL	localhost:3306 ssl	stud1_test	SQL	> desc customers;			
Field	Type	Null	Key	Default	Extra		
customerID	int	NO	PRI	NULL			
LastName	varchar(25)	YES		NULL			
FirstName	varchar(25)	YES		NULL			
address	varchar(100)	YES		NULL			
city	varchar(30)	YES		NULL			
email	varchar(40)	YES		NULL			
remark	varchar(300)	YES		NULL			
7 rows in set (0.0022 sec)							

Select * from customers;

MySQL	localhost:3306 ssl	stud1_test	SQL	> select * from customers;			
customerID	LastName	FirstName	address	city	email	remark	
5284	soni	kunal	daisy gardens	ambarnath	2025.kunal.soni@ves.ac.in	excellent customer	
12121	stan	mc	budhwar peth	p town	NULL	NULL	
12345	talokar	krishna	chaturbhuj	kharghar	NULL	NULL	
19725	leclerc	charles	st louis	monaco	NULL	NULL	
67698	devgn	praikr	press enclave	sion	NULL	NULL	
67890	soni	kunal	daisy gardens	ambarnath	2025.kunal.soni@ves.ac.in	excellent customer	
154382	kishan	ravi	parliament	amber	NULL	NULL	
174523	aubrey	drake	hehe huhu	toronto famm	NULL	NULL	
777777	thala	for a reason	railway quarters	ranchi	NULL	NULL	
8216784	karale	omi.r	idk mann	panvel	NULL	NULL	

select * from customers where customerid = 12345;

MySQL	localhost:3306 ssl	stud1_test	SQL	> select * from customers where customerid = 12345;			
customerID	LastName	FirstName	address	city	email	remark	
12345	talokar	krishna	chaturbhuj	kharghar	NULL	NULL	

insert into customers values

->123456,'doe','john','sample address','sample city','sample@gmail.com','sample customer');

```
MySQL localhost:3306 ssl stud1_test SQL > insert into customers values
-> (
-> 123456,'doe','john','sample address','sample city','sample@gmail.com','sample customer');
Query OK, 1 row affected (0.0062 sec)
```

select * from customers where customerid=123456;

```
MySQL localhost:3306 ssl stud1_test SQL > select * from customers where customerid=123456;
```

customerID	LastName	FirstName	address	city	email	remark
123456	doe	john	sample address	sample city	sample@gmail.com	sample customer

1 row in set (0.0008 sec)

Review Questions:

1. What is a database? Why do we need a database management system such as MySQL?

database is an organized collection of data that can be easily stored, accessed, and managed. A Database Management System (DBMS) such as MySQL is software used to create, store, modify, and retrieve data from databases. We need a DBMS because it helps us efficiently store large amounts of data, retrieve information quickly, update data, maintain accuracy, and manage data securely.

2. What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

CREATE DATABASE is used to create a new database in MySQL. SHOW DATABASES is used to display the list of databases available in MySQL. DROP DATABASE is used to permanently delete an existing database along with all the data stored in it.

3. What is a table in a database? Explain the purpose of rows and columns in a table.

A table is a structure in a database used to store data in an organized form. Columns represent the attributes or properties of the data, while rows represent individual records. For example, in a Customers table, CustomerID, Name, and Age can be columns, while each customer's information is stored in a separate row.

4. What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is a column that uniquely identifies each record in a table. CustomerID was selected as the primary key because every customer can have a unique CustomerID. This prevents duplicate identification of customers and makes it easy to find a particular customer's record.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

VARCHAR is a data type used to store variable-length text or character data. For example, Name VARCHAR(50) can be used to store a customer's name such as "Rahul". INT is a data type used to store whole numbers. For example, Age INT can be used to store values such as 18, 20, or 25.

6. What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

INSERT is used to add new records to a table. SELECT is used to retrieve or display data from a table. UPDATE is used to modify existing records in a table. ALTER TABLE is used to change the structure of an existing table, such as adding, deleting, or modifying columns.

7. Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The WHERE clause is important because it specifies which records should be updated. For example, `UPDATE Customers SET Age = 21 WHERE CustomerID = 101` will update only the customer whose CustomerID is 101. If an UPDATE statement is written without a WHERE condition, all records in the table will be updated. This can accidentally change the data of every customer and may result in incorrect or unwanted data.